Online Resource 2: Analytical Solution Derivation

Supporting Information for "When Inverse Physics-Informed Neural Networks Become Practically Identifiable: A Case Study in Synaptic Dopamine Transport"

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We derive the closed-form solution of the initial-boundary value problem of equations (1)–(4) of the main text on the unconfined planar domain $(x, y) \in \mathbb{R}^2$. This expression serves as the ground-truth reference for the forward-problem L^2 error reported in Table 2 of the main text.

The two-dimensional reaction-diffusion equation

$$\frac{\partial C}{\partial t} \;=\; D\,\nabla^2 C \;-\; k\,C, \quad \nabla^2 = \partial_{xx} + \partial_{yy}, \tag{1}$$

reduces to a pure diffusion equation under the substitution

$$C(x, y, t) \;=\; u(x, y, t)\,e^{-kt}, \tag{2}$$

which yields $\partial_t u = D\,\nabla^2 u$ for $u(x,y,t)$ and inherits the same radially symmetric initial profile $u(x,y,0) = C_0\,\exp\Big(-(x^2+y^2)/(2\sigma^2)\Big)$.

The fundamental solution of the diffusion equation on $\mathbb{R}^2$ is the two-dimensional heat kernel

$$G(x,y,t) \;=\; \frac{1}{4\pi D t}\,\exp\!\left(-\frac{x^2+y^2}{4Dt}\right), \tag{3}$$

and the solution with Gaussian initial data follows by two-dimensional convolution,

$$u(x,y,t) \;=\; \iint_{\mathbb{R}^2} G(x-x',\, y-y',\, t)\, u(x',y',0)\,\mathrm{d}x'\,\mathrm{d}y'. \tag{4}$$

Because both the heat kernel and the radially symmetric Gaussian initial condition separate as products of independent x and y factors, the integral evaluates to the product of two one-dimensional Gaussians,

$$u(x,y,t) \;=\; \frac{C_0\,\sigma^2}{\sigma^2+2Dt}\,\exp\!\left(-\frac{x^2+y^2}{2(\sigma^2+2Dt)}\right), \tag{5}$$

which describes a radially symmetric Gaussian whose variance grows linearly in time as $\sigma_t^2 = \sigma^2 + 2Dt$ in each Cartesian direction, while the integrated mass $\iint u\,\mathrm{d}x\,\mathrm{d}y = 2\pi\,C_0\,\sigma^2$ is conserved. Re-applying the substitution (2) gives the full analytical solution

$$\boxed{C(x,y,t) \;=\; \frac{C_0\,\sigma^2}{\sigma^2+2Dt}\,\exp\!\left(-\frac{x^2+y^2}{2(\sigma^2+2Dt)}\right)\,e^{-kt}.} \tag{6}$$

Equation (6) is exact on the unconfined domain $\mathbb{R}^2$. It satisfies the Neumann boundary conditions $\nabla C\cdot\hat{n} = 0$ on $\partial\Omega$ in the limit $L\to\infty$, and approximately when $\sigma_t \ll L/2$. For the parameter values used in this work ($L = 5\,\mu\mathrm{m}$, $\sigma = 0.5\,\mu\mathrm{m}$, $D = 0.32\,\mu\mathrm{m}^2/\mathrm{ms}$), this condition holds over the early portion of the simulation window; equation (6) therefore provides a reliable ground-truth reference in that regime, while finite-domain corrections via the method of images or a two-dimensional Fourier-cosine expansion become non-negligible at later times. The finite-difference reference solver of Section 3 of the main text serves as the complementary baseline that captures the boundary-driven deviations from (6) at large t.